TOMS Algorithms – Guidelines for Authors

v.1.0 (November 11, 2024) Paolo Bientinesi and Tim Hopkins

Introduction

A TOMS Algorithm submission consists of two files: A *pdf* file of the *Algorithm article*, and a *zipfile* containing the *software component*. The software component must contain all the software and data required to run all the example and test programs supplied as well as replicating any results presented in the accompanying article. It is not enough just to provide a link to an external repository like github; this is to ensure that reviewers all receive the same copy of the software, and that this material is not modified during the review process.

Both the article and the software will be subject to a formal reviewing process. Upon publication, the article will appear in a specific issue of TOMS within the ACM Digital Library and the accompanying software component will be made publicly available via ACM's CALGO (Collected Algorithms) web site (https://calgo.acm.org/).

Consult also: `https://dl.acm.org/journal/toms/algorithms-policy`

Checklist for Manuscript and Software

Please ensure that the following points are adhered to.

Manuscript

1. The title of the manuscript should be of the form:

   Algorithm XXXX: <Your title>

2. If you wish to include URLs to software repositories or other software components, add these as items to the bibliography and reference them from within the article rather than incorporating the URLs themselves within the text.

3. The same mechanism as in 2. should also be used to point the reader to where future versions of the software might be found; for example, *Future updated versions of the software package will be made available via [1].* Note: As mentioned above, an archival copy of the software will be made available via the CALGO website.

4. Use pseudo-code to define algorithms; these descriptions should be at a mathematical rather than an implementation language level. (An interested reader may consult the publicly available code if finer detail is required.)

5. Any claim that the submitted software is "better" than existing packages must be backed up with adequate comparisons.

6. When using colour in tables and figures, use a palette suitable for colour blind readers.

7. All words, numbers, and symbols in the figures must be easily readable. Ideally, their font size should be comparable to that of the caption for the figure. Use consistent fonts and font sizes across different figures.

8. Ensure all figures are referenced in the main text.

9. Avoid the use of footnotes as much as possible.

Software

1. Submit a single zip file containing the complete software component.

2. Organize the software so that the main directory contains only the file *userManual.pdf*, and the two directories *userManual/* and *Src/*. If the code includes multiple languages, in place of *Src/*, use one directory per language, and name them accordingly (for example, *C*, *CPP*, *Matlab*, *Python*, ...).

3. An optional third directory, *Doc/*, may be added at the top level to contain additional files that authors might like to include such as software licence files, top level *ReadMe* files, and similar.

4. Remove all files and directories such as *.git*, *.gitignore*, *.svn*, and any precompiled libraries (for example, **.a*) and object files (for example, **.o*).

5. Place all the source files needed to generate *userManual.pdf* in the top level directory *userManual/*. If the *pdf* file has been 'automatically' generated from, for example, comments in Python or R code, please provide both the LaTeX source files and any system dependent style files; the *pdf* file should be able to be regenerated using a standalone LaTeX system.

Code Specific

1. A submission consists of all the code and test data necessary for the effective use and testing of the algorithm.

2. The code must run (or be usable) at least under Linux and MacOS.

3. The code must contain drivers to replicate all results presented in the manuscript. (This is necessary for the badge, *Replicated Results*, to be awarded to the published Algorithm article.)

4. Files that allow a user to configure and build the package code in a portable way (for example, *make*, *cmake*, ...) must be provided. Similar files should be available for building and executing all the associated driver programs. Each driver program should be accompanied by any required data files (or data may be integrated into the program code, if desired) along with a file containing the expected output from executing the program. Supplied driver programs should *NOT* require interactive input.

5. (Not always possible): If comparisons are made with a competing package, it would be helpful if that package could also be included as part of the software component.

Algorithm Article vs User Manual

Algorithm papers describe specific algorithms and their implementation details, but they are not *user manuals*. In most cases, you want to ensure that the software that implements the algorithm is described at a high level in the Algorithm paper, but all details of the concrete use of the software must be described in the user manual (which is a part of the accompanying software component). Specifically, the following should appear in the manual, and *NOT* in the paper:

1. General implementation details along with a complete list of any dependent libraries or packages. Include the version numbers for which dependent software is known to support the author supplied code successfully.

2. All the details a user needs to install the software, and to run all the supplied driver and test programs.

3. Detailed descriptions of user callable procedures (i.e., subroutines, methods, functions, etc.), including descriptions of all user supplied arguments.

4. Illustrative code snippets showing, for example, how to solve a particular problem using the software.

5. Example usage of user callable procedures including sample outputs and, if applicable, the structure of any user supplied data files.

6. Advice on how to set any problem dependent parameters.